

Isaac Asimov's laws of robotics

Law zero. A robot may not injure humanity or, through inaction, allow humanity to come to harm.

First law. A robot may not injure a human being or, through inaction, allow a human being to come to harm.

Second law. A robot must obey orders given by human beings, except where such orders would conflict with the first law.

Third law. A robot must protect its own existence as long as such protection does not conflict with the first or second law.

Basic requirements of a robotic system

Some basic features of a robot include: **Mobility.** It possesses some form of mobility.

Programmability. It implies computational or symbol-manipulative capabilities that a designer can combine as desired. So the robot is basically a computer, which can be programmed to accomplish a large variety of tasks. After being programmed, it operates automatically.

Sensors. Sensors sense the environment and give useful feedback to the device.

Mechanical capability. Robots act according to the environment rather than merely functioning as a data processing or computational device.

Flexibility. Robots operate with a range of programs and manipulate and transport materials in a variety of ways.

Scope of robotics

Robotics requires the application of mechanical engineering, electrical/electronic engineering, computer-based integrated manufacturing, biological mechanics and software engineering.

A good depth in following major subjects is required to deal with robotics technology:

1. Artificial intelligence

Robotics, then and now

1801	A punch-card-controlled textile machine called Jacquard loom
1890	First autonomous vehicles (Tesla)
1922	First reference to 'robot' in Capek's play <i>Rossum's Universal Robots</i>
1938	Pollard and Roselund devised programmable paint sprayer
1946	DeVol—a general-purpose magnetic playback device for controlling machines
1946	Eckert & Mauchly devised ENIAC electronic computer
1948	MIT Prof. Wiener published <i>Cybernetics</i>
1952	First numerical control machine built at MIT
1954	Duval designed first programmable robot (Unimation—first robot company)
1959	Planet Corporation marketed the first commercially available robot
1962	GM installed first industrial robot on an assembly line
1964	Artificial Intelligence labs opened at MIT and SRI
1967	Mark II robot imported to Japan for paint spraying
1968	SRI built Shakey—first mobile robot operated using AI techniques
1970	First robot arm developed at Stanford
1973	T#—first minicomputer-controlled industrial robot (used extensively in the industry)
1976	Robot arms used on Viking 1 and 2 space probes
1977	ASEA (Europe) built first microcomputer-controlled robot
1980s	and on Rapid growth in the robot industry (Japan becomes the biggest player)

Source: <http://www.cs.binghamton.edu>

2. Engineering physics (mechanics)
3. Computer-aided design and computer-aided manufacturing
4. Computational geometry and simulation
5. Robot motion and path planning/detection
6. Electronic control, communication and navigation systems
7. Material science and technology
8. Optics engineering

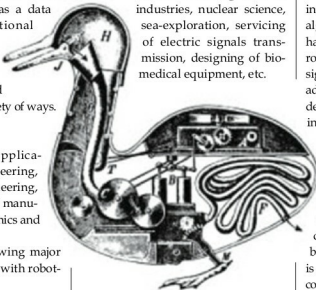
Robots are being used in industries, nuclear science, sea-exploration, servicing of electric signals transmission, designing of biomedical equipment, etc.

The interdisciplinary field of robotics also finds applications to pursue research in various sectors. A good background in mechanics, control and algorithms is the primary requirement for doing any work in robotics. In addition to this, if one would like to get involved in aerial robotics such as unmanned aircrafts, the basic knowledge of aerodynamics and aircraft mechanics is essential.

Robotics can be broadly divided into mechanical design and control algorithms. There is lot of research happening to improve the motion of robots through a good structural design and enhance their efficiency and adaptability through better control design, computer vision and artificial intelligence.

Challenges

Standardisation is a major hurdle. As no standard operating systems and software are currently in use, there exists diversity in the possible configurations in which a robot can be built and also in the hardware used. It is often necessary to write programming codes from scratch when dealing with new robots.



Vaucanson's digesting duck developed in 1739